

Course syllabus for

Academic year 2024/2025 >

FKA122 - Computational physics

Beräkningsfysik

Course syllabus adopted 2024-01-31 by Head of Programme (or corresponding)

Owner: [MPPHS](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: Second-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Course round 1

Teaching language: English

Application code: 85161

Open for exchange students: Yes

Block schedule: [D](#)

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course ¹	No Sp	Examination dates ¹
0124 Project 7,5 c Grading: TH		7,5 c					

In programs

[MPCAS COMPLEX ADAPTIVE SYSTEMS, MSC PROGR, Year 1 \(compulsory elective\)](#)

[MPCAS COMPLEX ADAPTIVE SYSTEMS, MSC PROGR, Year 2 \(elective\)](#)

[MPHPC HIGH-PERFORMANCE COMPUTER SYSTEMS, MSC PROGR, Year 1 \(elective\)](#)

[MPHPC HIGH-PERFORMANCE COMPUTER SYSTEMS, MSC PROGR, Year 2 \(elective\)](#)

[MPPHS PHYSICS, MSC PROGR, Year 1 \(compulsory elective\)](#)

Examiner:

[Julia Wiktor](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for Master's level (second cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Basic programming knowledge and experience, preferably in Python and C. Basic undergraduate physics.

Aim

The aim of the course is to refine computational skills by providing direct experience in using a computer to solve problems in physics. Numerical techniques are introduced and applied in a broad spectrum of various physical problems. The course is designed to develop an understanding of modeling physical systems using different numerical techniques.

Learning outcomes (after completion of the course the student should be able to)

- use C to solve numerical problems.
- explain and numerically apply the basic idea behind the molecular dynamics simulation method.
- explain how random numbers can be used to treat static and dynamic phenomena and numerically apply the methodology.

- explain and numerically apply the Metropolis Monte Carlo method.
- integrate knowledge in modeling physical systems with various numerical techniques.
- write well-structured technical reports where computational results are presented and explained.
- communicate results and conclusions in a clear way.

Content

- the programming language C
- ordinary differential equations, molecular dynamics simulation
- random numbers, random processes, Brownian dynamics
- discrete and fast Fourier transforms, power spectrum analysis
- Monte Carlo integration and the Metropolis algorithm
- Variational and diffusion Monte Carlo

Organisation

The different numerical techniques and the physical problems are presented in a series of lectures. The most important part in the course is the students own activity in applying the methods and solving a set of exercises and homework assignments. Scheduled computer laboratory sessions are provided, with instructors available for consultation. The programming language C is being used in the course.

Literature

Lecture notes will be made available.

Recommended additional literature.

For numerical methods:

William H. Press et al.,

"Numerical Recipes; The Art of Scientific Computing",
(3rd edition, Cambridge University Press, 2007),

For more experienced students:

J.M.Thijssen,

"Computational Physics",
(2nd edition, Cambridge University Press, 2007).

Examination including compulsory elements

The examination is based on computer exercises and project reports.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.